



2.05 Servicing the Vacuum Truck

1. Purpose

The purpose of the Standard Operating Procedure is to identify the hazards and define the steps in servicing the vacuum truck.

2. Scope

This procedure is intended for use by all Silverpoint employees.

3. Hazard Identification

- Physical Stress .
- Mental Stress
- Ergonomics .
- Physical Injury .
- Lifting , Bending .
- Twisting , Stretching .
- Slips , Trips and Falls .
- Hot Liquids .
- Cold Ground .

4. EHS Requirements

- Fire Retardant Coveralls .
- CSA Approved Steel Toed Boots .
- Safety Glasses .
- Rubber Gloves .
- Hard Hat .
- Hearing Protection .
- Follow Lock-Out Procedure .
- Take the Time to do Job Properly .

5. References



- Silverpoint Corporate Codes of Practice
- Silverpoint Task List
- Alberta OH&S
- Task List MED HAT THA 2.05

6. Procedure

- 6.1. Before you service or grease anything on vacuum /water truck you should turn the unit off and remove the key and set the brakes and place chock blocks at the wheels .
- 6.2. Every three days you must grease the truck to prevent excessive wear on equipment . If you divide the truck into 3 section and grease 1 section every day, it will shorten the amount of time in a day spent on that particular task . This way you are still greasing it every three days, but it is less of a time commitment when you do it.
- 6.3. It is important when greasing to check for excessive wear and defects in the equipment . This means physically taking hold of parts and checking them before greasing the part to ensure that the part is properly intact and there is no excessive movement . This will help to prevent breakdowns before they occur and possibly avoid incidents and injuries . Take note of the parts that might have oil leaking or metal shavings around them as well as other clues that would indicate that the part may be malfunctioning or perhaps broken .
- 6.4. In addition to this it is also crucial that if a grease nipple is not taking grease , it should be changed to ensure that the part is getting the grease that it needs , as well as the grease nipples that are pushing the grease back out of the hole after greasing , these would need to be changed as well .
- 6.5. On wells that require shift work then you will have to grease more often because the vacuum truck is worked harder . Record on PMI to properly communicate with co-workers as to the problems that need to be addressed . These items need to be recorded because they may be forgotten as the day progresses . If you identify a problem , do not feel you are at fault . **If it needs immediate attention , then call the shop or the 24 hours number .**

7. Maintenance :

- 7.1. Engine and Hibon oils are to be changed every 300 hrs and recorded in the PMI. Check the Transmission and Differentials a minimum of every two weeks , check for leaks everyday and record on your PMI and pre and post trip inspection . Check Engine and Antifreeze levels daily and report on your PMI and pre and post trip inspections . Transmission and differential oils are changed once a year in the shop .
- 7.2. **Greasing** : Doing a regular grease job on the truck is very important in the upkeep of the unit. Document this in your PMI, Hours and KM when you grease the truck. There are something's on the truck that should not be over greased , some of these are: TPC rotators and tank vibrator . The agitators that we have can not be over greased , 20 pumps min and should be done on a daily basis . When running the vibrator , you should never run for more than 30 seconds at a time .



Running them for long periods at a time can cause the welds to crack at the bottom of the tank. Water truck water pump should be greased every load.

- 7.3. To safely grease the rear door hinges, you will need to use the rig ladder. Using a spotter to steady the ladder, always use three points of contact and center yourself on the ladder at the required height and grease the door hinge as safely as possible. **DO NOT CLIMB ON TOP OF THE TRUCK**. You can also back up to a dirt pile and raise the tank until you can reach the grease nipples on the rear door hinge. You can do this while you are at the cement pit.
- 7.4. **Repairs** : When you come across an item that is in need of repairs, it is expected of you to repair what you can, call in for advice if needed. Often our work takes us to very remote parts of the province and dispatching a mechanic to your area may take some time. It can save a great deal of time and money when you could have simply repaired it yourself. This also applies to the shack. If there is a major problem, call the on-call number or your supervisor. It is also your responsibility to keep your shack and truck clean, inside and out. Some rigs will allow you to wash your truck on location, others will not, ask before you just start washing. If needed, you can phone in and possibly make arrangements to wash it at the closest town. Record in your PMI if any repairs need to be done or any repairs that you may have done to the truck or shack.
- 7.5. Once you are assigned a unit, you will be required to bring your own tools to aid you in repairs. Be sure to inventory your tools when you go on days off.
- 7.6. Some of the things that a person could check on a regular basis are:
 - 7.6.1. Air filters, in dusty conditions you could tap out the filter, find out where to pick up new filters closest to you. **Make sure the air filter goes back in the same way you took it out and that any gaskets or seals are properly replaced.**
 - 7.6.2. Brakes, slack adjusters, adjust brakes, record anything that doesn't look right and call in. Make sure that you do your tug test on the shack before starting a trip.
 - 7.6.3. Check lights and air hook ups as well. Make sure that you do your daily pre and post, and record anything that is broken.
 - 7.6.4. Checking your hydraulic level on your truck can indicate if you have a leak somewhere in the system, if the hydraulic level is down, check for oil leaks on the entire hydraulic system. If there are no visible leaks, it's possible the agitator lines in the vacuum tank could be leaking. Contamination in your hydraulic tank (contaminated hydraulic fluid looks whitish and milky) means that you have water or drilling mud in the system and is a serious problem you need to report immediately. You can also turn on your hydraulic fan manually to make sure that it is working, most of the trucks they will come on when the temperature gets up, but this way you know that it is working.
 - 7.6.5. Check oil and coolant every time before you start a cold engine
 - 7.6.6. Work lights are a big part of our work. Repair them if broken, keep them clean, point them where you can utilize them the most. It makes your job a lot easier when you have lights that work well. Make sure that you turn off your lights when on the main road; they are very bright and could cause an avoidable incident.
 - 7.6.7. Refer to SOP 1.03 for Agitator Maintenance.



- 7.6.8. Water truck : if your pump isn't sucking properly , you might need to remove a shim or two , off the end plates .
Call supervisor for help or advice .

7.7. TPC Repair Kit

- 7.7.1 . There is a TPC repair kit in every Vacuum truck , inside you will find a spare line to the wheels , extra fittings and a special wrench for replacing the rotary union .
- 7.7.2 . If a TPC line is damaged or tore off , it is quite easy to replace the line from the repair kit.

Remember : if you are in doubt about what to do, call your Supervisor